

Ransec – Empowering Cyber Defence using AI Powered Expert Systems

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Abstract: RANSec is an artificial intelligence cybersecurity framework designed to thwart ransomware threats in their earliest stages, thereby overcoming some of the deficiencies inherent in traditional signature-based security solutions. As ransomware evolves rapidly through polymorphic and zero-day variants, modern antivirus tools mostly fail to detect the threat before critical data becomes encrypted. To bridge this gap, RANSec uses various machine learning models-autoencoders, isolation forests, and random forests-to monitor system behavior in real-time, as well as reduce potential damage. The system will perfectly integrate with the set of established security tools like OSSEC and Snort for great and deep visibility and also for layer-based security. When the system detects a threat, RANSec can take action via quarantining files, killing malicious processes, isolating affected components, and notifying system administrators. This dual approach of AI-powered detection and rapid mitigation establishes a proactive, more scalable, and adaptive defense mechanism that can protect diverse digital environments from emerging ransomware threats.

Key Words: Ransomware Detection, Cybersecurity, Machine Learning, Autoencoder, Isolation Forest, Random Forest, Anomaly Detection, Automated Threat Mitigation.

1. **INTRODUCTION:** Ransomware has now started to be considered one of the most disrupting and hazardous varieties of cyber-attacks, capable of locking up critical data, paralyzing operations, and ultimately causing severe financial loss. With attack techniques developing fast, including polymorphic behavior and zero-day variants, ransomware can easily bypass traditional signature-based antivirus systems. Most security tools today detect threats after the malicious encryption has already begun, which leaves organizations with very little time for response and recovery. This increasing sophistication underlines the urgent need for proactive and intelligent defense mechanisms that will detect abnormal behavior before any actual damage has been done.

Going ahead to address the challenge, AI-powered cybersecurity systems are becoming a necessity for modern digital environments. These apply machine learning models and real-time behavioral analysis to detect subtle anomalies that may point toward an early stage of a ransomware attack. Applying AI with system monitors and established security tools allows for faster detection, automated responses, and better decision-making. This project is focused on such an advanced defense system, not only to enhance the accuracy of detection but also to minimize downtime and prevent large-scale data loss.

2. **LITERATURE REVIEW:** The integration of sensors, automation, and IoT in agriculture has been explored in several research studies.

Ransomware Detection Based on Artificial Intelligence and Machine Learning: A Review (2025) the authors proposed a hybrid approach, in which anomaly-detection models such as autoencoders and isolation forest can be combined with supervised classifiers like SVM and Random Forest. This system analyzed log activity, network behavior, and file-metadata features and showed that early-stage ransomware activities were better detected by this approach.

A Comprehensive Literature Review on Ransomware (2024) A state-of-the-art survey of machine learning and deep learning techniques in ransomware detection was presented. The work summarized the current state of leading models, datasets, and challenges; that keeps the scientific community moving from signature-based tools to intelligent behavioral detection methods.

Ransomware Detection Using Machine Learning Techniques (2024) Comparing different supervised learning models, including decision trees, SVM, and multi-layer perceptron. Results proved that decision trees yielded high accuracy (~98.8%) on ransomware datasets, thus proving their efficiency for classification tasks in traditional ML models.

Ransomware Detection using Machine Learning: A Survey by 2023 reviewed the trends, algorithms, and developed detection strategies within the period between 2017 and 2022. The study has highlighted the progress of automated detection pipelines and the ever-increasing relevance of behavioral features for identifying new ransomware variants.

Dynamic Feature Dataset for Ransomware Detection Using Machine Learning (2023) The developed dynamic runtime behavioral dataset captured features like file, process, and system-call. Results of machine learning classifiers, trained with this dataset, showed successful ransomware detection for both the categories, encryptor-type and locker-type, with high performance.

Ransomware Detection and Classification using Machine Learning (2023) Employed Random Forest and XGBoost models to classify ransomware families based on behavioral patterns. The study demonstrated high precision while emphasizing the importance of feature engineering in model robustness.

Ransomware Detection Based on Memory Dumps (2024) focused on the detection of unknown ransomware by feature analysis of memory dumps combined with machine learning. The proposed model showed a high accuracy result, representing the value of memory-based behavioral analysis.

Improving Cybersecurity: Comparative Insights in Machine Learning Models for Ransomware Detection, 2025 The various ML models evaluated in the study included SVM, Random Forest, ANN, and Gradient Boosting. Each of the classifiers presented strengths and weaknesses; the study thereby recommended hybrid approaches.

Toward Effective Machine Learning Models for Ransomware Detection (2024) Assessed various supervised models using features derived from time-series measures of ransomware activity. These results show that these temporal patterns of behaviour significantly improve the early detectability of ransomware attacks.

API Call Analysis with Explainable Machine Learning (2022)-API call sequences dataset of 15 ransomware families, and applied ML models with SHAP-based explainability. This empowered the system to identify ransomware with transparent reasoning of its predictions.

3. **OBJECTIVES / AIMS:** The The RANsec system aims to build an intelligent, proactive cybersecurity defense against ransomware through a hybrid AI engine. It accomplishes real-time threat detection by analyzing anomalies in system activities. Its principal objective is to apply automated mitigation responses, such as quarantining files and terminating processes, so that threats are swiftly contained before widespread encryption of valuable data can happen. The system is designed with easy integration with other open-sourced security tools like OSSEC and Snort to further enhance overall security visibility. Lastly, it claims cross-platform compatibility--Mac, Windows, Linux--and an intuitive web dashboard for user monitoring and configuration.

4. RESEARCH METHOD / METHODOLOGY:

- a. **Monitoring Layer :** This layer is in charge of continuously collecting data on system activity. It incorporates external security technologies such as Snort IDS/PS and OSSEC Agents. In particular, the File Monitor (Watcher) pushes event details (path, size, extension, timestamp) to an event queue and keeps track of all file system activities (create, edit, delete, rename).
- b. **Layer of Detection**
Feature Extraction: Based on the observed events, the system calculates behavioral features such the number of writes during the last 10 seconds, the number of modified unique file extensions, and the ratio of extension changes.
AI Engine (Hybrid Model): A hybrid AI engine receives the extracted information and uses them to make predictions. This engine detects anomalies suggestive of ransomware using a variety of machine learning models:
Isolation Forest: To identify anomalous file modifications. An autoencoder is a neural network that detects deviations based on high reconstruction error after learning typical behavior.
Random Forest: For supervised classification, determining whether an action is ransomware or not.
Rule Engine: This part executes actions depending on the outputs of the AI model using non-machine learning, rule- based reasoning (e.g., IF anomaly_score > 0.7 THEN quarantine file).
- c. **Layer of Response :** The Response Handler immediately initiates automated activities upon detecting a threat, Termination of Process: Malicious processes are stopped,
Suspicious files are transferred to a quarantine folder using Quarantine Manager. Network Isolation: The impacted system can be cut off from the network.
Alerting System: The administrator receives notifications through text, SMS, or email.
- d. **Layer of Visualization :** Administrators can view system status, a list of recent alerts, set monitoring paths, adjust thresholds, and verify true or false positives via a Web Dashboard.

5. FINDINGS / RESULTS:

The RANSec system successfully demonstrated a proactive and automated defense mechanism against simulated ransomware threats. Key findings and outputs include:

•**Real-time Threat Detection:** The hybrid AI engine, combining Isolation Forest, Autoencoder, and Random Forest, successfully monitors file, process, and network activity to detect anomalous behaviors characteristic of ransomware in real-time.

•**Automated Response & Mitigation:** Upon detecting a threat, the system immediately executed automated mitigation actions. Console outputs demonstrated that the malicious process was successfully terminated and an alert was generated, preventing further execution of the simulated ransomware.

•**Multi-Channel Alerting:** The system provided instant notifications through multiple channels, ensuring the administrator was immediately alerted to the security breach. Outputs showed successful alerts delivered via Email (Gmail) and SMS/Text (using Twilio).

•**Layered Visibility:** RANSec achieved layered security visibility by integrating with existing tools like OSSEC and Snort, validating the objective of seamless integration.

•**Intuitive Monitoring Dashboard:** The system included a fully functional Web Dashboard for visualizing the system status, active alerts, and threat logs, providing an intuitive interface for administrators.

•**Proactive Defense:** The AI-driven approach and automated prevention capability allow RANSec to mitigate damage before widespread data encryption occurs, addressing the limitations of traditional, signature-based tools.

6. **DISCUSSION / ANALYSIS:** RANsec is the next logical step in cybersecurity—a realm exceeding the limited capacity of traditional signature-based tools that are becoming totally ineffective against advanced zero-day and polymorphic ransomware variants. The Hybrid AI engine at its heart combines a mix of supervised-unsupervised machine learning models, including Random Forest, Isolation Forest, and Autoencoder, to establish a set baseline for normal system behavior and then detect in real-time anomalies in file, process, and network activities. By means of this proactive behavioral detection capability, the system is empowered to identify such threats rapidly, almost before serious harm could be caused by extended encryption. The embedded rules engine will also ensure that high-confidence predictions from the ML models will automatically trigger a response very rapidly and consistently, including file quarantine, terminate malicious processes, and network isolation. Besides, its easy integration into well-established security platforms like OSSEC and Snort assures that it enhances and protects existing security infrastructure without demanding a complete overhaul, which makes it a robust, scalable, and adaptive digital defense for modern environments.
7. **RECOMMENDATIONS:** Integrate a self-learning AI engine, add a feature for automatic rollback of file changes and recovering the system to a pre-infection state, ensure full, optimized compatibility across all target environments: Mac, Windows and Linux, develop advanced pre-attack simulations.
8. **CONCLUSION / SUMMARY:** The RANsec system offers an extendable proactive solution to deal with the never-ending evolution of ransomware threats, which traditional signature-based tools normally fail to detect, particularly against zero-day and polymorphic variants. In fact, the power of this system is rooted in its AI-driven expert system, which relies on a hybrid model comprising machine learning algorithms (autoencoders, isolation forests, and random forests) that monitor system behavior in real time and categorize any anomalies as an ongoing ransomware attack. This behavioral-intelligent approach makes it possible to detect ransomware well before a large part of the data is encrypted, thereby giving a unique advantage over existing approaches that are often late in warning about this. By combining continuous system monitoring with automated response capabilities, RANSec significantly reduces the window of opportunity for attackers. It achieves this with a structured architecture comprising a monitoring layer, using OSSEC and Snort; a detection layer, which is the AI engine; a response layer, which will handle process termination, file quarantine, and network isolation; and a visualization layer, normally a web dashboard. With this design, rapid detection along with automated prevention ensures that the system can quickly contain threats and alert the administrators through multi-channel notifications, such as email and SMS. Finally, RANSec stands out as an efficient, intelligent, and adaptive defense mechanism that keeps the potential damage at its lowest and provides protection for a wide range of digital environments.
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